



Barker
College

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Student Number

2022

TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION

Mathematics Advanced

Staff Involved:

• ALY • PCB • RJW • RAS • DXC*
• ESP • AXD • GPF • JZT

Friday August 5th, 2022

160 copies

General

Instructions:

- Reading time - 10 minutes
- Writing time - 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided separately
- For questions in Section II, show relevant mathematical reasoning and / or calculations
- Marks may not be awarded for careless or poorly arranged working
- Diagrams are not to scale unless specifically stated

Total marks:

100

Section I - 10 marks (pages 2 - 6)

- Attempt Multiple Choice Questions 1 - 10
- Allow about 20 minutes for this section

Section II - 90 marks (pages 7 - 27)

- Attempt Questions 11 - 33
- Show all necessary working
- Allow about 2 hours and 40 minutes for this section

Section I

10 marks

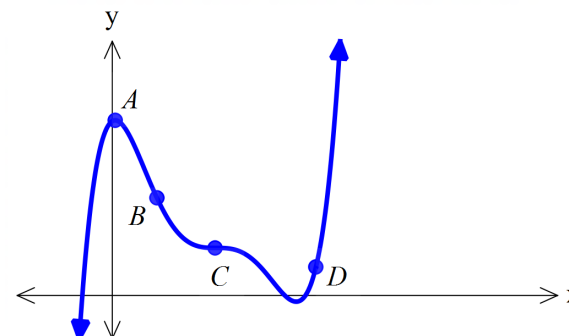
Attempt Questions 1 – 10

Allow about 20 minutes for this section

1 What is domain of the function $y = \ln(2x - 3)$?

- A. All real x
- B. $(0, \infty)$
- C. $\left[\frac{3}{2}, \infty\right)$
- D. $\left(\frac{3}{2}, \infty\right)$

2 At which point on this curve are **both** the first derivative and second derivative equal to zero?



- A. A
- B. B
- C. C
- D. D

- 3 Evaluate the following integral:

$$\int_{-\frac{\pi}{6}}^{\frac{\pi}{3}} (\cos x + 1) dx$$

- A. $\frac{\sqrt{3}}{2} + \frac{1}{2} + \frac{\pi}{2}$
 B. $\frac{\sqrt{3}}{2} - \frac{1}{2} + \frac{\pi}{2}$
 C. $\frac{\sqrt{3}}{2} + \frac{1}{2} + \frac{\pi}{6}$
 D. $\frac{\sqrt{3}}{2} - \frac{1}{2} - \frac{\pi}{6}$

- 4 Which of the following equations is the reflection of the curve $y = 3^x + 2$ in the x -axis, then translated 2 units downwards?

- A. $y = 3^{-x}$
 B. $y = 3^{-x} - 4$
 C. $y = -3^x$
 D. $y = -3^x - 4$

- 5 What values of x satisfy the equation $|3x - 4| - 2 = 3$?

- A. $x = -3$ and $x = \frac{1}{3}$
 B. $x = -\frac{1}{3}$ and $x = 3$
 C. $x = 1$ and $x = 3$
 D. $x = -1$ and $x = \frac{1}{3}$

- 6 There are 16 T-shirts in a basket. 10 of the T-shirts are blue and 8 of the T-shirts have stripes. 4 of the T-shirts are neither blue nor have stripes. What is the probability of taking a T-shirt at random from the basket that is blue and has stripes?

- A. $\frac{1}{8}$
 B. $\frac{1}{4}$
 C. $\frac{3}{8}$
 D. $\frac{1}{2}$

- 7 Which of the following is equal to $\frac{\ln 8^x + 3 \ln 3^x}{2 \ln 3 + \ln 8 - \ln 2}$?

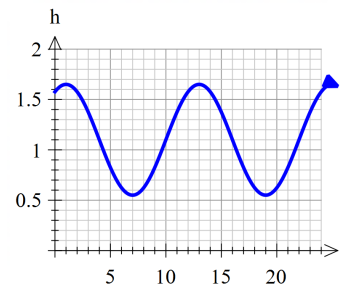
- A. x
 B. $\frac{3x}{2}$
 C. $3x$
 D. $6x$

- 8 The height of the water in a harbor is modelled by the following function:

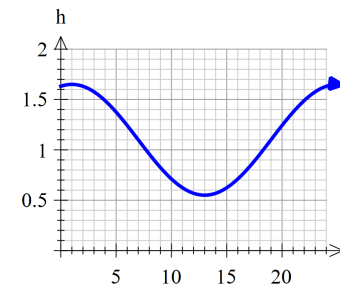
$$h(t) = 0.55 \cos\left(\frac{\pi}{6}(t - 1)\right) + 1.1$$

Which of the following is the graph of this function?

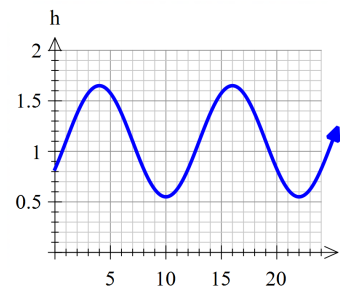
A.



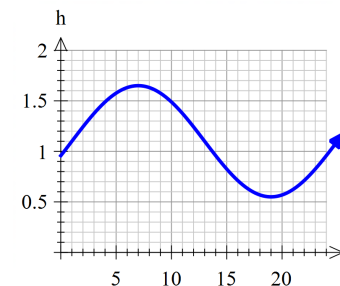
B.



C.



D.



- 9 The probability density function for the continuous random variable X is:

$$f(x) = \begin{cases} 2xe^{-x^2} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

What is the mode of the continuous random variable X ?

- A. $\ln 2$
- B. $\sqrt{\ln 2}$
- C. $\sqrt{2}$
- D. $\frac{1}{\sqrt{2}}$

- 10 $f(x)$ is an odd function defined over all real x .

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It is known that $\int_0^1 f(x) \, dx = 2$ and $\int_{-1}^0 f(x) \, dx = \int_1^3 f(x) \, dx$.

What is the value of $\int_0^4 f(x-1) \, dx$?

- A. -2
- B. 0
- C. 2
- D. 4

End of Section I

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Student Number

Section II

90 marks

Attempt Questions 11 – 33

Allow about 2 hours and 40 minutes for this section.

Write your student number on all answer pages.

In Questions 11 – 33, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (1 mark)

Differentiate $e^x \cos x$.

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Question 12 (1 mark)

Find $\int 3^x dx$

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Question 13 (2 marks)

Evaluate the arithmetic series $3 + 6.5 + 10 + 13.5 + \dots + 66$

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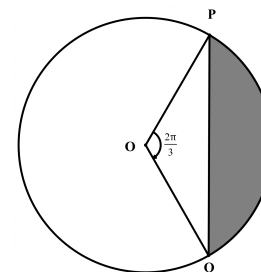
Question 14 (2 marks)

The diagram shows a circle with centre O .

The points P and Q are lying on the circle such that $\angle POQ = \frac{2\pi}{3}$.

If the triangle POQ has an area of $8\sqrt{3} \text{ cm}^2$, find the exact area of the shaded minor segment.

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Question 15 (3 marks)

Find the equation of the normal to the curve $y = x^4 - 6x - 3$ at the point on the curve where $x = 2$.

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Student Number

Question 16 (4 marks)

In the mobile game *Desmos Royale*, players earn 0, 1, 2, or 3 star points after completing every level of the game. Let the discrete random variable X be the number of star points that a player earns from completing a level in the game.

x	0	1	2	3
$p(X = x)$	0.1	$2a$	$5a$	a

- (a) Find the value of a . 1

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- (b) Calculate the expected value of X . 2

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- (c) Unlocking a feature in the game costs 50 star points. How many levels should a player be expected to complete in order to earn enough star points to unlock a feature? 1

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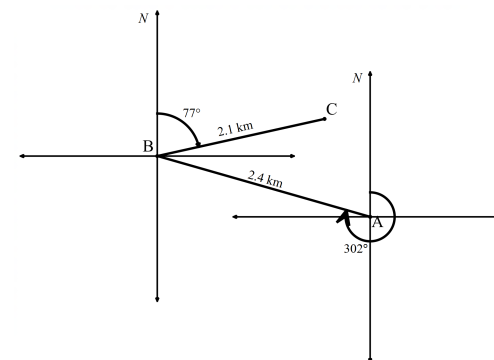
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Question 17 (3 marks)

Ms Kalnins flies a drone for 2.4 km on a bearing of 302° from location A to location B . She then flies the drone for 2.1 km on a bearing of 077° to location C .

The information is shown in the diagram below.



- (a) What is the size of $\angle ABC$? 1

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- (b) What is the distance from point A to point C ?
Give your answer correct to 1 decimal place. 2

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Question 18 (2 marks)

A pencil case contains 7 blue pens and 3 red pens. Two pens are selected at random without replacement. What is the probability of selecting at least one red pen?

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Student Number

Question 19 (5 marks)

The number N of E. coli bacteria that is present in a petri dish after t minutes is given by the following equation:

$$N = 6e^{kt}$$

The population of E. coli bacteria will double in number in 20 minutes.

- (a) Find the exact value of k . 2

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- (b) Determine the exact number of bacteria after 6 hours. 1

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- (c) At what rate is the number of bacteria increasing after 6 hours?
Give your answer correct to the nearest whole number. 2

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Question 20 (4 marks)

Consider the graph of $f(x) = 3 \sin\left(2x - \frac{\pi}{3}\right)$

- (a) State the amplitude and period for the graph. 2

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- (b) Solve the equation $3 \sin\left(2x - \frac{\pi}{3}\right) = 3$ over the domain $0 \leq x \leq \pi$. 2

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Question 22 (2 marks)

If $f'(x) = 3xe^{x^2}$, find $f(x)$ if the graph $y = f(x)$ has a y -intercept of 4. **2**

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Question 23 (2 marks)

A fuel pump connected to a power generator operates at a varying rate, $R(t)$, for an hour, **2**
 where $R(t)$ is in L/min. The rate is recorded every 10 minutes and is given in the table below.

t	0	10	20	30	40	50	60
$R(t)$	264	295	264	250	268	273	286

The number of litres of fuel pumped over this time frame is given by

$$\int_0^{60} R(t) dt$$

Using the trapezoidal rule with 7 function values, find the approximate number of litres of fuel pumped over the course of the hour.

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Question 24 (5 marks)

A film director invites 24 film critics to rate his latest film out of 5 stars. The table below shows the ratings given by the film critics:

Number of stars	Frequency
0	3
1	4
2	5
3	7
4	3
5	2

(a) Calculate the mean and median for this data set. **2**

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The director decided to record the gender of each film critic along with their review score in a two-way table:

	Male	Female	Total
3 or more stars			
Less than 3 stars		9	
Total	11		

(b) Using the information stated previously, complete the rest of the above two-way table. **2**

(c) What is the probability a film critic is male, given the critic gave the film a score of 3 or more stars? **1**

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Question 25 (2 marks)

Find the point(s) of intersection of the graphs of $y = 9^x + 4$ and $y = 13 - 8 \times 3^x$. **2**

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Question 26 (3 marks)

2.4L of organic material is added to a compost bin every day. Each day, bacteria in the compost bin break down 12.5% of the total material in the compost bin. Initially, the compost bin is empty.

- (a) How much organic material is in the compost bin immediately after material is added on the second day? **1**

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- (b) The maximum capacity of the compost bin is 18L. Is the capacity of the bin sufficient for its current usage? Give justification as to why or why not. **2**

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Question 27 (4 marks)

The average Sydney daily temperature, $T(x)$, in degrees Celsius, is modelled by the equation

$$T(x) = 22 + 4 \cos\left(\frac{2\pi x}{366}\right) \quad \text{for } 0 \leq x \leq 365$$

where x is the number of days after 1st January in a leap year.

- (a) State the average Sydney daily temperature to 2 decimal places for February 25th. **1**

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- (b) Find the values of x for when the average Sydney daily temperature is 20°C. **3**

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Question 29 (5 marks)

- (a) Show that $\frac{d}{dx} \left(\ln \left(\frac{x}{x+1} \right) \right) = \frac{1}{x^2+x}$ 2

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- (b) Hence, or otherwise, find the exact value of k if $\int_1^3 \frac{2}{x^2+x} dx = \ln k$. 3

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Question 30 (4 marks)

A particle is initially at the origin and its velocity v m/sec after t seconds is given by:

$$v = 2t(2t - 1)(t - 1)$$

- (a) Find the displacement function, x , and the times when the particle is at the origin. 2

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- (b) Calculate the total distance the particle has travelled in the first second. 2

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$$f(x) = \begin{cases} kx(1-x)^2 & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

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2

$$F(x) = \begin{cases} \rule{1.5cm}{0.4pt}, & \text{for } x < 0 \\ \rule{1.5cm}{0.4pt}, & \text{for } 0 \leq x \leq 1 \\ \rule{1.5cm}{0.4pt}, & \text{for } x > 1 \end{cases}$$

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$$E(X) = \int_a^b xf(x) dx$$

$$E(X^2) = \int_a^b x^2 f(x) dx$$

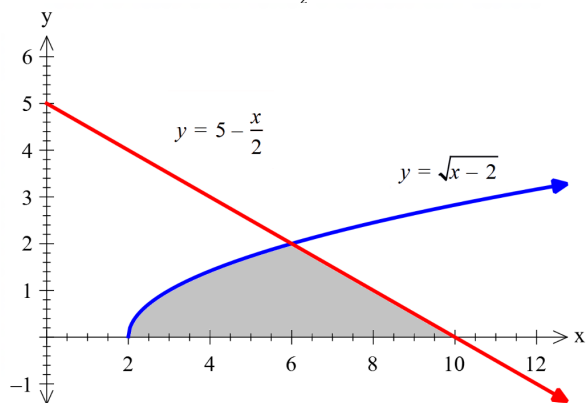
Hence, calculate the mean and standard deviation of the probability distribution.

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Question 32 (3 marks)

The diagram below shows the graphs of $y = 5 - \frac{x}{2}$ and $y = \sqrt{x - 2}$.



- (a) Show that there is a point of intersection when $x = 6$.

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- (b) Find the exact value of the shaded area in simplest form.

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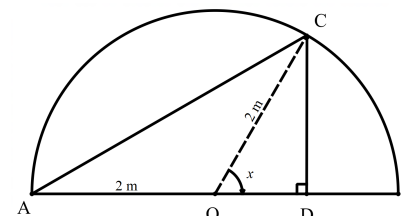
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Question 33 (6 marks)

Let A and B form the diameter of a semicircle of radius 2 metres and centre at O. From a point D lying on the line AB, a perpendicular is drawn meeting the circumference of the semicircle at C. Let $\angle COD = x$.



- (a) Show that the area, $A(x)$, of triangle ACD can be given by the equation:

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$$A(x) = 2 \sin x (1 + \cos x)$$

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- (b) Hence, find the value of x that gives triangle ACD the greatest area.

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Section II extra writing space

If you use this space, clearly indicate what question you are answering.

1.) $2x - 3 > 0$

$$2x > 3$$

$$x > \frac{3}{2}$$

$$\therefore x \in \left(\frac{3}{2}, \infty\right)$$

(D)

2.) Point C is the only point where the gradient of the tangent = 0 ($y' = 0$) AND there is a point of inflexion. ($y'' = 0$)

(C)

$$\begin{aligned} 3.) \int_{-\frac{\pi}{6}}^{\frac{\pi}{3}} (\cos x + 1) dx &= \left[\sin x + x \right]_{-\frac{\pi}{6}}^{\frac{\pi}{3}} \\ &= \left(\sin \frac{\pi}{3} + \frac{\pi}{3} \right) - \left(\sin \left(-\frac{\pi}{6}\right) + \left(-\frac{\pi}{6}\right) \right) \\ &= \frac{\sqrt{3}}{2} + \frac{\pi}{3} + \frac{1}{2} + \frac{\pi}{6} \\ &= \frac{\sqrt{3}}{2} + \frac{1}{2} + \frac{\pi}{2} \end{aligned}$$

(A)

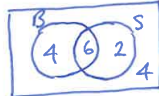
4.) $y = 3^x + 2$
Reflection about x-axis $\rightarrow y = -(3^x + 2)$
Translation 2 units down $\rightarrow y = -3^x - 2 - 2 = -3^x - 4$

(D)

$$\begin{aligned} 5.) |3x - 4| - 2 &= 3 \\ |3x - 4| &= 5 \\ 3x - 4 &= 5 \text{ or } 3x - 4 = -5 \\ 3x &= 9 & 3x &= -1 \\ x &= 3 & x &= -\frac{1}{3} \end{aligned}$$

(B)

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6.)  Blue or striped = 16 - 4 = 12
Blue and striped = 10 + 8 - 12 = 6
 $P(\text{Blue and striped}) = \frac{6}{16} = \frac{3}{8}$

(C)

$$\begin{aligned} 7.) \frac{\ln(8^x) + 3\ln(3^x)}{2\ln 3 + \ln 8 - \ln 2} &= \frac{\ln(2^{3x}) + 3x\ln 3}{2\ln 3 + \ln 2^3 - \ln 2} \\ &= \frac{3x\ln 2 + 3x\ln 3}{2\ln 3 + 3\ln 2 - \ln 2} \\ &= \frac{3x(\ln 2 + \ln 3)}{2\ln 3 + 2\ln 2} \\ &= \frac{3x(\ln 2 + \ln 3)}{2(\ln 3 + \ln 2)} \\ &= \frac{3x}{2} \end{aligned}$$

(B)

8.) Period = $\frac{2\pi}{\frac{\pi}{6}} = 12$

Only (B) and (C) have a period of 12.
Horizontal shift 1 to the right.
Curve curve is symmetrical about y-axis, so a 2 unit shift to right results in (A)

9.) Mode occurs when $f(x)$ peaks $\rightarrow f'(x) = 0$
 $f'(x) = 2e^{-x^2} - 4x^2e^{-x^2}$
when $2e^{-x^2} - 4x^2e^{-x^2} = 0$
 $2e^{-x^2}(1 - 2x^2) = 0$
 $1 - 2x^2 = 0$
 $2x^2 = 1$
 $x^2 = \frac{1}{2}$
 $x = \pm \frac{1}{\sqrt{2}}$, but $x \geq 0$
 $\therefore$ (D)

$$10.) \int_0^4 f(x-1) dx = \int_{-1}^3 f(x) dx$$

$$= \int_{-1}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^3 f(x) dx$$

Since $f(x)$ is odd: $\int_{-1}^0 f(x) dx = -\int_0^1 f(x) dx = -2$

$$\therefore \text{integral} = -2 + 2 - 2 = -2$$

(A)

$$11.) \frac{d}{dx}(e^x \cos x) = e^x \cos x + e^x(-\sin x)$$

$$= e^x \cos x - e^x \sin x$$

$$12.) \text{From the reference sheet: } 3^x = e^{x \ln 3}$$

$$\therefore \int 3^x dx = \int e^{x \ln 3} dx$$

$$= \frac{e^{x \ln 3}}{\ln 3} + C$$

$$= \frac{3^x}{\ln 3} + C$$

13.) Arithmetic Series:

$$a=3 \quad d=3.5$$

$$T_n = 3 + 3.5(n-1)$$

$$\text{when } T_n = 66$$

$$66 = 3 + 3.5(n-1)$$

$$63 = 3.5(n-1)$$

$$18 = n-1$$

$$\therefore n = 19$$

$$\therefore S_{19} = \frac{19}{2}(3+66)$$

$$= 655.5$$

$$14.) \text{Area } \triangle POQ = \frac{1}{2} r^2 \sin \theta$$

$$8\sqrt{3} = \frac{1}{2} r^2 \sin \frac{2\pi}{3}$$

$$8\sqrt{3} = \frac{1}{2} r^2 \left(\frac{\sqrt{3}}{2}\right)$$

$$\therefore r^2 = 32$$

$$\text{Area of segment} = \frac{1}{2} r^2 (\theta - \sin \theta)$$

$$= \frac{1}{2} \times 32 \left(\frac{2\pi}{3} - \sin \frac{2\pi}{3}\right)$$

$$= 16 \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2}\right)$$

$$= \left(\frac{32\pi}{3} - 8\sqrt{3}\right) \text{ cm}^2$$

$$15.) y' = 4x^3 - 6$$

when $x=2$ grad. of tangent $= 4 \times 2^3 - 6 = 26$

$$\therefore \text{grad. of normal} = -\frac{1}{26}$$

Egⁿ of normal

$$m = -\frac{1}{26}$$

$$x=2$$

$$y = 2^4 - 6 \times 2 - 3$$

$$= 1$$

$$y-1 = -\frac{1}{26}(x-2)$$

$$y-1 = -\frac{x}{26} + \frac{1}{13}$$

$$y = \frac{x}{26} + \frac{14}{13} \quad \text{OR} \quad x + 26y - 28 = 0$$

$$16a.) 0.1 + 2a + 5a + a = 1$$

$$8a = 0.9$$

$$a = \frac{0.9}{8}$$

$$= \frac{9}{80}$$

$$b.) E(x) = 0 \times 0.1 + 1 \times 2a + 2 \times 5a + 3 \times a$$

$$= 2a + 10a + 3a$$

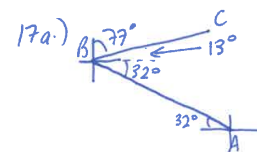
$$= 15a$$

$$= 15 \times \frac{9}{80}$$

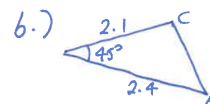
$$= 1.6875$$

$$c.) \frac{50}{1.6875} = 29.6 \dots$$

$\therefore$ need to complete 30 levels



$$\angle ABC = 32^\circ + 13^\circ = 45^\circ$$



$$AC^2 = 2.1^2 + 2.4^2 - 2(2.1)(2.4)\cos 45^\circ$$

$$\therefore AC = 1.7 \text{ km (1.d.p.)}$$

$$18.) P(\text{at least 1 red}) = 1 - P(\text{no red})$$

$$= 1 - \frac{7}{10} \times \frac{6}{9}$$

$$= \frac{8}{15}$$

$$19.) a.) \text{when } t=0$$

$$N=6$$

$$\text{when } t=20$$

$$N=12 \rightarrow \text{sub into } N=6e^{kt}$$

$$12 = 6e^{20k}$$

$$2 = e^{20k}$$

$$20k = \ln 2$$

$$k = \frac{\ln 2}{20}$$

b.) doubles every 20 min.

$$6 \text{ hours} = 360 \text{ min.}$$

$$\therefore \text{no. of bacteria doubles } \frac{360}{20} = 18 \text{ times.}$$

After 6 hours, there are:

$$6 \times 2^{18} = 1572864$$

$$19c.) \frac{dN}{dt} = 6ke^{kt}$$

$$= k \times 6e^{kt}$$

$$\text{when } t=360, 6e^{kt} = 1572864$$

$$\therefore \frac{dN}{dt} = \frac{\ln 2}{20} \times 1572864$$

$$= 54511 \text{ bacteria/min. (nearest whole)}$$

$$20a.) A=3$$

$$A = \frac{2\pi}{2} = \pi$$

$$b.) 3 \sin(2x - \frac{\pi}{3}) = 3, \quad 0 \leq x \leq \pi$$

$$0 \leq 2x \leq 2\pi$$

$$-\frac{\pi}{3} \leq 2x - \frac{\pi}{3} \leq 2\pi - \frac{\pi}{3}$$

$$\sin(2x - \frac{\pi}{3}) = 1$$

$$2x - \frac{\pi}{3} = \frac{\pi}{2} \quad \text{in this domain, only solution is } \frac{\pi}{2}$$

$$2x = \frac{\pi}{2} + \frac{\pi}{3}$$

$$2x = \frac{5\pi}{6}$$

$$x = \frac{5\pi}{12}$$

$$21a.) f(x) = x^3 + 5x^2 - 8x - 40$$

$$\text{when } f(x) = 0$$

$$x^3 + 5x^2 - 8x - 40 = 0$$

$$x^2(x+5) - 8(x+5) = 0$$

$$(x+5)(x^2 - 8) = 0$$

$$x = -5 \quad x^2 = 8$$

$$x = \pm 2\sqrt{2}$$

$$\therefore x\text{-ints are } x = -5, 2\sqrt{2}, -2\sqrt{2}$$

$$b.) f'(x) = 3x^2 + 10x - 8$$

$$\text{when } f'(x) = 0$$

$$3x^2 + 10x - 8 = 0$$

$$(3x-2)(x+4) = 0$$

$$x = \frac{2}{3}, \quad x = -4$$

$$y = \frac{-1156}{27}, \quad y = 8$$

21b.) (continued)

Suspected stationary points are:

$$\left(\frac{2}{3}, -\frac{1156}{27}\right) \text{ and } (-4, 8)$$

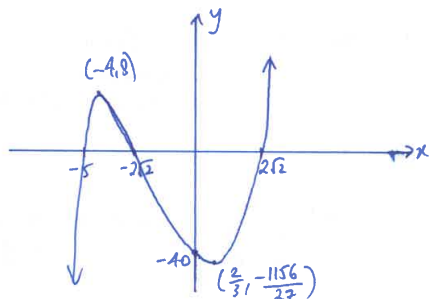
Testing using $f''(x)$:

$$f''(x) = 6x + 10$$

$$f''\left(\frac{2}{3}\right) = 14 > 0 \therefore \left(\frac{2}{3}, -\frac{1156}{27}\right) \text{ is a local min T.P.}$$

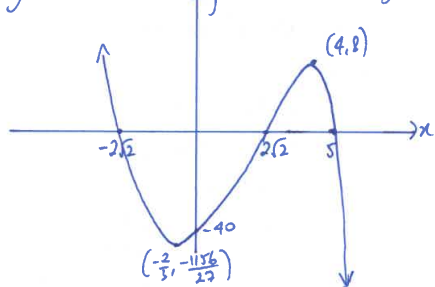
$$f''(-4) = -14 < 0 \therefore (-4, 8) \text{ is a local max T.P.}$$

21c.)

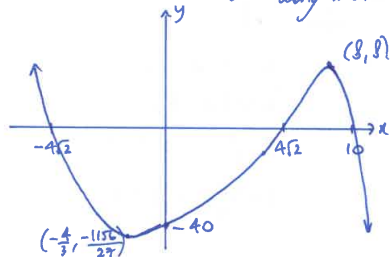


21d.) $y = f(-x)$

reflection about y-axis



$y = f\left(-\frac{x}{2}\right) \rightarrow$ dilation by a factor of 2 along x-axis



$$\begin{aligned} 22.) f(x) &= \int 3xe^{x^2} dx \\ &= \frac{3}{2} \int 2xe^{x^2} dx \\ &= \frac{3}{2} e^{x^2} + C \end{aligned}$$

$$\text{when } x=0, y=4$$

$$4 = \frac{3}{2} e^0 + C$$

$$4 = \frac{3}{2} + C$$

$$\therefore C = \frac{5}{2}$$

$$\therefore f(x) = \frac{3}{2} e^{x^2} + \frac{5}{2}$$

23.)

$$\begin{aligned} \int_0^{60} R(t) dt &\approx \frac{10}{2} \left[264 + 286 \right. \\ &\quad \left. + 2(295 + 264 + 250 + 268 + 233) \right] \\ &= 16260 \text{ L} \end{aligned}$$

24.) a)

$$\bar{x} = \frac{0 \times 3 + 1 \times 4 + 2 \times 5 + 3 \times 7 + 4 \times 3 + 5 \times 2}{24}$$

$$= 2.375$$

median = halfway between 12th & 13th score

$$= 2.5$$

b.)

	M	F	Total
≥ 3	d	b	c
< 3	e	q	f
Total	11	a	g

$$11 + a = 24$$

$$a = 13$$

$$b + q = 13$$

$$b = 4$$

$$c = \text{frequency of three or more stars} = 12$$

$$d + 4 = 12$$

$$d = 8$$

$$d + e = 11$$

$$e = 3$$

$$f = 3 + q$$

$$= 12$$

$$g = 12 + 12 = 24$$

$$\begin{aligned} 24c.) P(\text{male} | 3 \text{ or more stars}) &= \frac{8}{12} \\ &= \frac{2}{3} \end{aligned}$$

$$25.) \text{ when } 9^x + 4 = 13 - 8(3^x)$$

$$(3^x)^x + 4 = 13 - 8(3^x)$$

$$(3^x)^2 + 4 = 13 - 8(3^x)$$

$$(3^x)^2 + 8(3^x) - 9 = 0$$

$$(3^x - 1)(3^x + 9) = 0$$

$$3^x = 1, \quad 3^x = -9$$

$$x = 0 \quad \text{? rejected}$$

$$\begin{aligned} \text{when } x=0 \\ y &= 9^0 + 4 \\ &= 5 \end{aligned}$$

$$(0, 5)$$

26a.) Let A_n = amount in bin after n^{th} addition

$$A_1 = 2.4$$

$$A_2 = 0.875 \times A_1 + 2.4$$

$$= 0.875 \times 2.4 + 2.4$$

$$= 4.5 \text{ L}$$

$$b.) A_3 = 0.875 \times A_2 + 2.4$$

$$= 0.875^2 \times 2.4 + 0.875 \times 2.4 + 2.4$$

$$A_n = 2.4 \times 0.875^{n-1} + \dots + 2.4$$

$$\text{since } |r| < 1$$

$$A_{\infty} = \frac{a}{1-r}$$

$$= \frac{2.4}{0.125}$$

$$= 19.2 \text{ L}$$

$\therefore$ The capacity of the bin is NOT sufficient for how much compost it would eventually need to contain.

$$\begin{aligned} 27.) a.) \text{ no. of days paired} &= 30 + 25 = 55 \\ \text{when } x = 55 \quad T(55) &= 22 + 4 \cos\left(\frac{2\pi \times 55}{366}\right) \\ &= 24.35^\circ \text{C (2.d.p.)} \end{aligned}$$

b.) Let x when $T(x) = 20$

$$20 = 22 + 4 \cos\left(\frac{2\pi x}{366}\right), \quad 0 \leq \frac{2\pi x}{366} \leq \frac{2\pi \times 365}{366}$$

$$-2 = 4 \cos\left(\frac{2\pi x}{366}\right)$$

$$\cos\left(\frac{2\pi x}{366}\right) = -\frac{1}{2}$$

Looking for solutions in 2nd and 3rd quad.
 the upper limit of the domain is close to 2π

$$\frac{2\pi x}{366} = \frac{2\pi}{3}, \frac{4\pi}{3}$$

$$\frac{x}{366} = \frac{1}{3}, \frac{2}{3}$$

$$x = 122, 244$$

28a.) monthly rate = $\frac{4.8\%}{12} = 0.4\% = 0.004$

$$A_1 = 532000(1.004)^1 - M$$

$$A_2 = 1.004 \times A_1 - M$$

$$= 532000(1.004)^2 - M(1.004) - M$$

$$\vdots$$

$$A_n = 532000(1.004)^n - M(1.004)^{n-1} - \dots - M$$

$$= 532000(1.004)^n - M(1.004^{n-1} + \dots + 1)$$

$$= 532000(1.004)^n - M\left(\frac{1.004^n - 1}{1.004 - 1}\right)$$

$$= 532000(1.004)^n - M\left(\frac{1.004^n - 1}{0.004}\right)$$

$$= 532000(1.004)^n - 250M(1.004^n - 1)$$

b.) when $n = 25 \times 12 = 300, A_{300} = 0$

$$0 = 532000(1.004)^{300} - 250M(1.004^{300} - 1)$$

$$250M(1.004^{300} - 1) = 532000(1.004)^{300}$$

$$M = \frac{532000(1.004)^{300}}{250(1.004^{300} - 1)}$$

$$= \$3048.34$$

28c.) Without 20% deposit:

$$\text{Total repayment} = 3048.34 \times 300 = 914502$$

$$\text{Total interest paid} = 914502 - 532000 = 382502$$

With 20% deposit:

$$\text{Amount borrowed} = 0.8 \times 532000 = 425600$$

$$\text{Monthly repayment} = 0.8 \times 3048.34 = 2438.672$$

$$\therefore \text{Total repayment} = 2438.672 \times 300 = 731601.60$$

$$\text{Total interest paid} = 731601.60 - 425600 = 306001.60$$

$$\text{Saving in interest} = 382502 - 306001.60 = 76500.40$$

29a.) Derivative of inner function:

$$\frac{d\left(\frac{x}{x+1}\right)}{dx} = \frac{(x+1)(1) - x(1)}{(x+1)^2}$$

$$= \frac{1}{(x+1)^2}$$

$$\frac{d(\ln(f(x)))}{dx} = \frac{f'(x)}{f(x)}$$

$$\therefore \frac{d(\ln(\frac{x}{x+1}))}{dx} = \frac{1}{(x+1)^2} \div \left(\frac{x}{x+1}\right)$$

$$= \frac{1}{(x+1)^2} \times \frac{(x+1)}{x}$$

$$= \frac{1}{x(x+1)}$$

$$= \frac{1}{x^2+x}$$

$$29b.) \int_1^3 \frac{2}{x+x} dx = \ln k$$

$$2 \int_1^3 \frac{1}{x+x} dx = \ln k$$

$$2 \left[\ln\left(\frac{x}{x+1}\right) \right]_1^3 = \ln k$$

$$2 \left[\ln\left(\frac{3}{4}\right) - \ln\left(\frac{1}{2}\right) \right] = \ln k$$

$$2 \left(\ln\left(\frac{3}{4} \div \frac{1}{2}\right) \right) = \ln k$$

$$2 \left(\ln\left(\frac{3}{2}\right) \right) = \ln k$$

$$\ln\left(\frac{9}{4}\right) = \ln k$$

$$\therefore k = \frac{9}{4}$$

$$30a.) v = 2t(2t-1)(t-1)$$

$$= (4t^2 - 2t)(t-1)$$

$$= 4t^3 - 6t^2 + 2t$$

$$x = \int (4t^3 - 6t^2 + 2t) dt$$

$$= t^4 - 2t^3 + t^2 + C$$

$$\text{when } t=0, x=0 \therefore C=0$$

$$\therefore x = t^4 - 2t^3 + t^2$$

$$\text{when } x=0$$

$$t^4 - 2t^3 + t^2 = 0$$

$$t^2(t^2 - 2t + 1) = 0$$

$$t^2(t-1)^2 = 0$$

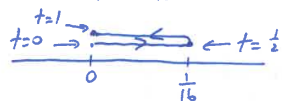
$$t=0, t=1$$

$$30b.) \text{ when } v=0$$

$$2t(2t-1)(t-1) = 0$$

$$t=0, t=\frac{1}{2}, t=1$$

$$x=0, x=\frac{1}{16}, x=0$$



$$\therefore \text{total dist. travelled} = 2 \times \frac{1}{16} = \frac{1}{8} \text{ metre}$$

$$31a.) \int_0^1 f(x) dx = 1$$

$$k \int_0^1 x(1-x)^2 dx = 1$$

$$k \int_0^1 x(1-2x+x^2) dx = 1$$

$$k \int_0^1 (x - 2x^2 + x^3) dx = 1$$

$$k \left[\frac{x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{4} \right]_0^1 = 1$$

$$k \left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4} - 0 \right) = 1$$

$$k \times \frac{1}{12} = 1$$

$$k = 12$$

$$b.) F(x) = \int_0^x 12t(1-t)^2 dt = 12 \left(\frac{x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{4} \right) \text{ (homogeneous part)}$$

$$F(0) = 0$$

$$F(1) = 1$$

$$\therefore F(x) = \begin{cases} 0 & x < 0 \\ 12\left(\frac{x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{4}\right) & 0 \leq x \leq 1 \\ 1 & x > 1 \end{cases}$$

$$31c.) P(X \geq 0.5) = 1 - P(X \leq 0.5)$$

$$= 1 - F(0.5)$$

$$= 1 - 12 \left(\frac{0.5^2}{2} - \frac{2 \times 0.5^3}{3} + \frac{0.5^4}{4} \right)$$

$$= 0.3125 \text{ or } 31.25\%$$

$$31d.) E(X) = \int_0^1 x^2(1-x)^2 dx$$

$$= \int_0^1 (x^2 - 2x^3 + x^4) dx$$

$$= 12 \left[\frac{x^3}{3} - \frac{2x^4}{4} + \frac{x^5}{5} \right]_0^1$$

$$= 12 \left(\frac{1}{3} - \frac{1}{2} + \frac{1}{5} - 0 \right) = \frac{2}{5} \text{ or } 0.4$$

$$\begin{aligned} E(X^2) &= 12 \int_0^1 x^3(1-x)^2 dx \\ &= 12 \int_0^1 (x^3 - 2x^4 + x^5) dx \\ &= 12 \left[\frac{x^4}{4} - \frac{2x^5}{5} + \frac{x^6}{6} \right]_0^1 \\ &= 12 \left(\frac{1}{4} - \frac{2}{5} + \frac{1}{6} \right) \\ &= 0.2 \text{ or } \frac{1}{5} \end{aligned}$$

$$\begin{aligned} \text{var}(X) &= E(X^2) - [E(X)]^2 \\ &= 0.2 - \left(\frac{1}{5}\right)^2 \\ &= 0.04 \\ \therefore \text{s.d.} &= \sqrt{0.04} \\ &= 0.2 \end{aligned}$$

$$32a.) \text{ when } x=6$$

$$y_1 = 5 - \frac{6}{2} = 2, \quad y_2 = \sqrt{6-2} = 2$$

$$\therefore y_1 = y_2 \text{ and point of int. occurs at } x=6$$

$$\begin{aligned} b.) \text{ Area} &= \int_2^6 \sqrt{x-2} dx + \text{area of a triangle} \\ &= \int_2^6 \sqrt{x-2} dx + \frac{1}{2}(10-6) \times 2 \\ &= \int_2^6 (x-2)^{\frac{1}{2}} dx + 4 \\ &= \left[\frac{2}{3}(x-2)^{\frac{3}{2}} \right]_2^6 + 4 \\ &= \frac{2}{3}(4^{\frac{3}{2}} - 0) + 4 \\ &= \frac{2}{3} \times 8 + 4 \\ &= \frac{28}{3} \text{ u}^2 \end{aligned}$$

33a.) In $\triangle OCO$:

$$\cos x = \frac{OD}{2} \Rightarrow OD = 2\cos x$$

$$\sin x = \frac{CD}{2} \Rightarrow CD = 2\sin x$$

$$\begin{aligned} A(x) &= \frac{1}{2} \times AD \times CD \\ &= \frac{1}{2} (2 + 2\cos x) (2\sin x) \\ &= 2\sin x (1 + \cos x) \end{aligned}$$

$$\begin{aligned} \text{b.) } A'(x) &= 2\cos x (1 + \cos x) + 2\sin x (-\sin x) \\ &= 2\cos x + 2\cos^2 x - 2\sin^2 x \\ &= 2\cos x + 2\cos^2 x - 2(1 - \cos^2 x) \\ &= 4\cos^2 x + 2\cos x - 2 \end{aligned}$$

when $A'(x) = 0$

$$4\cos^2 x + 2\cos x - 2 = 0$$

$$2\cos^2 x + \cos x - 1 = 0$$

$$(2\cos x - 1)(\cos x + 1) = 0$$

$$\cos x = \frac{1}{2} \quad \cos x = -1$$

$$x = \frac{\pi}{3} \quad \text{no solution since } x < \frac{\pi}{2}$$

test:

x	0	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$A'(x)$	+	-	-ve

$$\begin{aligned} A'(0) &= 4 + 2 - 2 \\ &= 4 \\ &> 0 \end{aligned}$$

$$\begin{aligned} A'(\frac{\pi}{2}) &= 0 + 0 - 2 \\ &< 0 \end{aligned}$$

$$\therefore \text{max at } x = \frac{\pi}{3}$$

$$\therefore A(x) \text{ is greatest when } x = \frac{\pi}{3}$$